Kriging method and its modifications

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Presentation Outline

Background: Pixel-based Classification in GIS

Context: Remote Sensing and Expectations Maximization (EM) Clustering

- High-resolution data demands automated, pixel-wise Land Use/Land Cover (LULC) classification or clustering.
- The EM clustering algorithm was used in the data used in this study, and five clustering results were given.

Tobler's 1st Law

"Everything is related to everything else, but near things are more related than distant things." [4]

The Weakness of Expectation-Maximization Algorithm: EM Clustering treat every single pixel as an independent sample. this directly violates Tobler's First Law of Geography. Natural landscapes are continuous and spatially autocorrelated.

Problem Statement: The 'Salt-and-Pepper' Effect

The Phenomenon: This happens when individual pixels, or small, fragmented groups, stand out against their background. Their labels, as assigned by the EM algorithm, clash with the surrounding data, creating a noisy, scattered appearance that doesn't match the environment. [1].

Consequences in GIS

- Reduced-quality LULC decision-making.
- Inaccurate spatial area estimation.
- Damaged landscape ecology metrics.

Limitations of Heuristic Filters

- **Majority Filter:** Purely geometric, removes thin linear features (e.g., roads, rivers) [3].
- **Markov Random Fields (MRF):** Tends to over-smooth real geographic boundaries.
- **Object-Based Image Analysis (OBIA):** Relies heavily on subjective parameters.

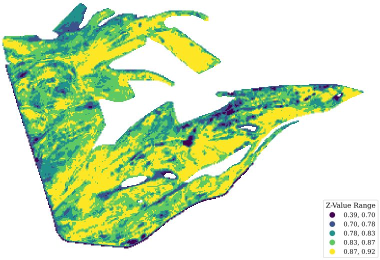
Research Objectives & Pipeline

The Core Objective

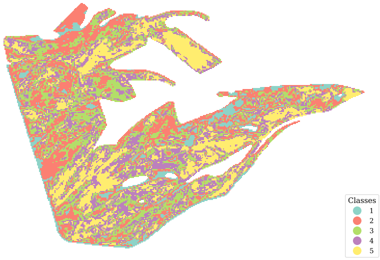
Correct isolated labels so they make sense in the context of the surrounding physical landscape (Z).

The Workflow:

1. **Identify isolated spots:** Use spatial connectivity metrics to flag pixels that stand out from their neighbors.
2. **Model the expected value:** Use Kriging to predict what the physical value should be at those locations.
3. **Test the consistency:** Check if the current labels actually match the underlying physical data.
4. **Reassign labels:** Apply Maximum Likelihood Estimation to reassign structurally inconsistent labels.



(a) Productivity Distribution (Z)



(b) Fragmented Categorical Map (L)

Here, the continuous physical attribute Z represents a real-valued scalar corresponding to the maximum seasonal NDVI, and the categorical L represents agricultural productivity levels.

- The continuous physical field (a) initially shows spatial autocorrelation.
- The categorical map (b) shows high-frequency noise.

Verification of Prerequisite: Spatial Autocorrelation

Fundamental Assumption

Ordinary Kriging requires the attribute field $Z(\mathbf{X})$ to show spatial autocorrelation, rather than complete spatial randomness [2].

Quantitative Assessment (Global Moran's I)

$$I = \frac{N}{W} \frac{\sum_i \sum_j w_{ij}(Z_i - \bar{Z})(Z_j - \bar{Z})}{\sum_i (Z_i - \bar{Z})^2} \tag{1}$$

- **Result:** A highly significant positive spatial autocorrelation ($I = 0.8002$, $p < 0.001$).
- **Implication:** This provides the theoretical support to model the physical field $Z(\mathbf{X})$ as a **Spatial Random Field (SRF)** and apply Kriging methods.

Data Normalization

Normality Violation: Original data is apparently non-Normal distributed. This violates the normality assumption in later hypothesis testing, which will lead to biased predictions and invalid hypothesis testing [5].

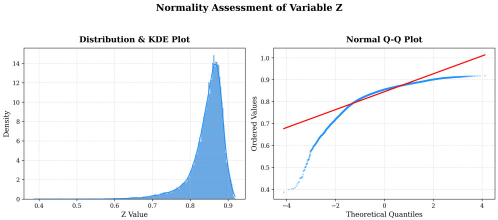
Normal Score Transformation (NST)

$$Y(\mathbf{X}_i) = \Phi^{-1}(F(Z(\mathbf{X}_i))) \qquad (2)$$

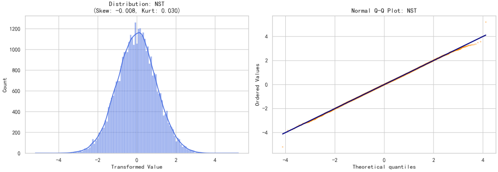
** Φ^{-1}: Inverse standard normal CDF; F: Cumulative Distribution Function of Z.*

Transformation Results:

- Regularized spatial skewness to ≈ 0.
- The empirical quantiles align with the Gaussian diagonal (see Q-Q plot).



(a) Original Z Histogram & Q-Q Plot



(b) NST Transformed Y Histogram & Q-Q Plot

Step 1: Isolation Detection (Candidate Extraction)

Objective: Identify spatially isolated pixels, which are likely misclassified rather than being true geographical features.

Mathematical Metrics

- **Neighbor Disagreement Score (NDS):** Measures spatial mismatch.

$$NDS(\mathbf{X}_i) = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \mathbb{I}\Big(L(\mathbf{X}_j) \neq L(\mathbf{X}_i)\Big) \tag{3}$$

  ** $\mathbb{I}(\cdot)$ is the indicator function; $\mathcal{N}_i$ defines the Moore neighborhood.*

- **Connected Components Analysis (CCA):** Evaluates spatial connectivity to flag clusters below the set Minimum Unit.

Algorithmic Logic: Denote $\mathcal{S}$ as the set of all pixels, $\mathcal{S}_{\text{NDS}}$ as the set of pixels with $NDS > \tau$ (e.g., $\tau = 0.8$), and $\mathcal{S}_{\text{CCA}}$ as the set of pixels in connected components smaller than the Minimum Unit (e.g., 4 pixels). The pixels within the union set $\mathcal{S}_{\text{cand}} = \mathcal{S}_{\text{NDS}} \cup \mathcal{S}_{\text{CCA}}$ are considered isolated and serve as prediction targets in the subsequent Kriging stage.

Step 2: Geostatistical Validation (Ordinary Kriging)

For candidate $\mathbf{X}_0 \in \mathcal{S}_{\text{cand}}$, we use Ordinary Kriging to predict the expected value $\hat{Z}(\mathbf{X}_0)$ and measure the associated uncertainty $\sigma_K^2(\mathbf{X}_0)$.

Kriging Methodology

$$\textbf{Estimator:}\ \ \hat{Z}(\mathbf{X}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{X}_i), \quad \mathbf{X_i} \in \mathcal{S}_{\text{reference}} = \mathcal{S} \backslash \mathcal{S}_{\text{cand}} \tag{4}$$

$$\textbf{Variance:}\ \ \sigma_K^2(\mathbf{X}_0) = 2\sum_{i=1}^{n} \lambda_i \gamma(\mathbf{X}_i, \mathbf{X}_0) - \sum_{i=1}^{n}\sum_{j=1}^{n} \lambda_i \lambda_j \gamma(\mathbf{X}_i, \mathbf{X}_j) \tag{5}$$

λ_i are the Kriging weights determined by solving the Kriging system of equations, γ is the semivariogram to model spatial dependence.

Why Ordinary Kriging?

- Best Linear Unbiased Predictor.
- Provides physically grounded uncertainty σ_K^2.

Computational Optimization: The computation complexity is $\mathcal{O}(N^3)$, therefore uses only reference pixels that are within a certain distance from the candidate pixel.

Step 3: Hypothesis Testing Framework

- H_0 **(The "Normal" case):** The observed value at this location is consistent with the surrounding area. It looks like something that would naturally occur given its neighbors.
- H_A **(The "Outlier" case):** The observed value is out of place. It seems to come from a different process, meaning it doesn't fit in with the nearby environment.

Test Statistic $\eta(\mathbf{X}_0)$:

$$\eta(\mathbf{X}_0) = \frac{Z_{\text{obs}}(\mathbf{X}_0) - \hat{Z}(\mathbf{X}_0)}{\sigma(\mathbf{X}_0)} \sim \mathcal{N}(0,1) \tag{6}$$

** Asymptotic normality is mathematically guaranteed by the prior NST step.*

To avoid the multiple testing problem across all candidates, we apply the Benjamini-Hochberg procedure to control the False Discovery Rate (FDR) at a pre-specified level $q = 0.05$.

Decision Rule

1. **Reject H_0:** Large physical difference $\rightarrow$ **Keep original label** (Real feature).
2. **Fail to reject H_0:** Small difference $\rightarrow$ **Proceed to Reassignment** (Noise).

Step 4: Likelihood-Based Class Reassignment

Kriging-Based Likelihood Function

For each candidate class $C_k \in \mathcal{C}$, we compute the likelihood of observing $Z_{\text{obs}}(\mathbf{X}_0)$:

$$\mathcal{L}(C_k, \mathbf{X}_0) = \frac{1}{\sigma_{C_k}(\mathbf{X}_0)} \exp\left(-\frac{(Z_{\text{obs}}(\mathbf{X}_0) - \hat{Z}_{C_k}(\mathbf{X}_0))^2}{2\sigma^2_{C_k}(\mathbf{X}_0)}\right) \tag{7}$$

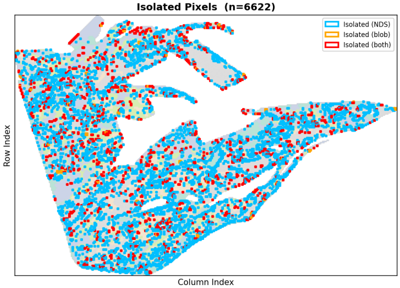
** $\hat{Z}_{C_k}(\mathbf{X}_0)$ and $\sigma^2_{C_k}(\mathbf{X}_0)$ are the Kriging prediction and variance computed using only the reference pixels belonging to class C_k, i.e., $\mathcal{S}_{ref}(C_k) = \{\mathbf{X}_i : L(\mathbf{X}_i) = C_k\}$.*

- **Structural Regularization:** The leading term $\sigma^{-1}_{C_k}$ mathematically penalizes candidate classes that show high spatial uncertainty or limited local data.
- The corrected final label L^* is chosen to maximize the likelihood function across all candidate classes $\mathcal{C}$ (i.e., all 5 classes):

$$L^*(\mathbf{X}_0) = \underset{C_k \in \mathcal{C}}{\arg\max}\ \mathcal{L}(C_k, \mathbf{X}_0)$$

Result 1: Extraction of Candidate Outliers

(a) Continuous NDS Distribution



(b) Extracted Spatial Candidates

Observations from Step 1:

- **NDS Map (a):** High spatial mismatches are primarily clustered along class boundaries.
- **Candidates (b):** Effectively isolates the "salt-and-pepper" noise patches. **6,622 pixels** flagged as candidates ($\approx 18.4\%$ of total domain).

Result 2: Semivariogram Modeling

Based on the weighted Root Mean Square Error (wRMSE), the Exponential model demonstrated the most optimal fit.

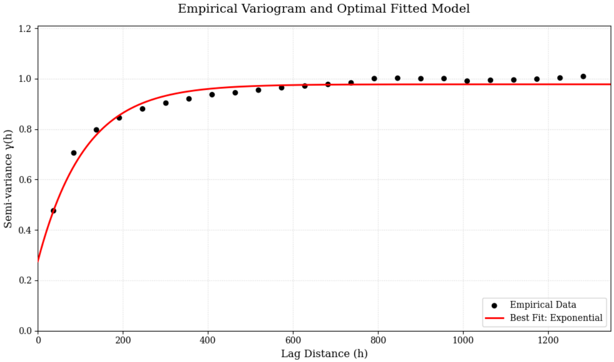
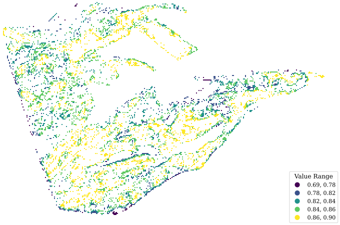


Figure 4: Empirical Semivariogram (black dots) and Fitted Models: Exponential (red).

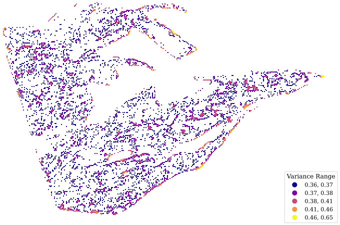
Fitted Parameters:

- **Nugget (c_0):** 0.275, reflects inherent sensor/environmental noise.
- **Sill (c):** 0.978, indicates the total variance of the spatial process.
- **Range (a):** 110.0 meters, defines the spatial correlation distance.

Result 3 & 4: Kriging Validation & Statistical Decisions



(a) Kriging Prediction Field ($\hat{Z}$)



(b) Kriging Variance Field (σ_K^2)

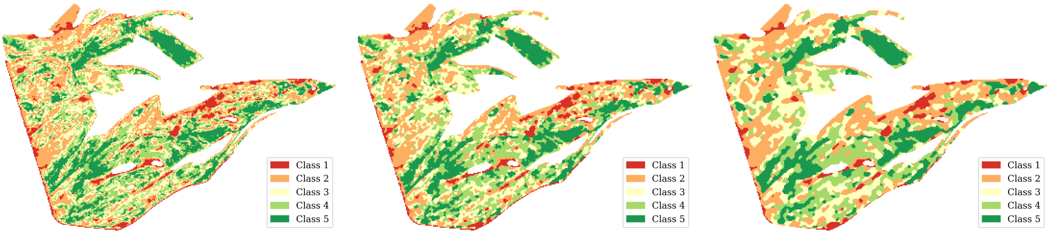
- **Prediction (a):** Smooth continuous field behind categorical noise.
- **Variance (b):** Uncertainty spatially measured ($\sigma_K^2 \in [0.36, 0.65]$); high variance zones guide hypothesis testing.

FDR Control at $q = 0.05$:

Statistical Category	Count	percentage, %
Total Candidates (S_{cand})	6,622	100%
Rejected H_0	**6**	**0.1%**
Fail to reject H_0	6,616	99.9%

- 99.9% of the "salt-and-pepper" noise considered as noise pixels.
- FDR test keeps **6 real micro-structures** that standard filters would have removed.

Result 5: Label Correction Analysis and Comparison



(a) Original Fragmented Map (b) Refined Map (c) Majority Filtered Map

- The Kriging-based refinement (b) reassigns 5,180 pixels, retaining 1,436 pixels, which are original micro-features. It is a blance between noise removal and feature preservation.
- In contrast, the Majority Filter (c) reassigns 9,542 pixels and over-smooths the map.

Conclusion & Future Directions

Research Summary:

- Transitioned post-classification from "geometric cleanup" to **formal spatial inference**.
- Successfully balanced strong noise removal with the keeping of micro-scale geographic realities.

Future Research Directions:

- Replace OK with **Universal Kriging** or **Regression Kriging** to include additional variables (e.g., DEM, Soil Moisture).
- Expanding $\gamma(\mathbf{h})$ into **3D semivariograms** $\gamma(\mathbf{h}, t)$ to validate pixels against their historical physical paths.
- Using cross-correlation between multiple spectral indices to improve refinement stability.

References I

[1] Julian Besag. "On the statistical analysis of dirty pictures". In: *Journal of the Royal Statistical Society: Series B (Methodological)* 48.3 (1986), pp. 259–279.

[2] Noel Cressie. *Statistics for spatial data*. John Wiley & Sons, 2015.

[3] Kwang E Kim. "Adaptive majority filtering for contextual classification of remote sensing data". In: *International Journal of Remote Sensing* 17.5 (1996), pp. 1083–1087.

[4] Waldo R Tobler. "A computer movie simulating urban growth in the Detroit region". In: *Economic geography* 46.2 (1970), pp. 234–240.

[5] Hans Wackernagel. *Multivariate geostatistics: an introduction with applications*. Springer Science & Business Media, 2003.

Thank You!

Questions & Discussion